13

Sensors, Electrodes, and Transducers

13–1 OBJECTIVES

1. To learn the principles behind the operation of common types of transducers.
2. To learn how to specify and apply transducers.
3. To learn the limitations of certain transducers.
4. To learn about certain application problems and how they are solved.

13–2 SELF-EVALUATION

Before studying the material in this chapter, try to answer the questions given below. These questions test your knowledge of the subject matter. If you cannot answer a particular question, then look for the answer as you read the text.

1. Describe the differences between **bonded** and **unbonded** piezoresistive strain gages.
2. List three different types of temperature transducers.
3. List two types of gas flow transducers.
4. Define **transducer** in your own words.
5. How may a force be measured if only displacement transducers are available?

13–3 TRANSDUCERS AND TRANSDUCTION

Not all of the physical variables that must be measured lend themselves to direct input into electronic instruments and circuits. Unfortunately, electronic circuits operate only with inputs that are currents and voltages. So, when one is measuring nonelectrical physical quantities, it becomes necessary to provide a device that converts physical parameters such as *force, displacement, temperature,* and so forth, into proportional voltages or currents. The **transducer** is such a device.

> **Definition:** A **transducer** is a device or apparatus that converts nonelectrical physical parameters into electrical signals (i.e., currents or voltages) that are proportional to the value of the physical parameter being measured.

Transducers take many forms and may be based on a wide variety of physical phenomena. Even when one is measuring the *same* parameter, different instruments may use different types of transducers.

This chapter will not be an exhaustive catalogue treatment covering all transducers—manufacturers' data sheets may be used for that purpose—but we will discuss some of the more common *types* of transducers used in scientific, industrial, medical, and engineering applications.

13–4 THE WHEATSTONE BRIDGE

Many forms of transducers create a variation in an electrical resistance, inductance, or capacitance in response to some physical parameter. These transducers are often in the form of a Wheatstone bridge or one of the related ac bridge circuits. In many cases where the transducer itself is not in the form of a bridge, it is used in a bridge circuit with other components forming the other arms of the bridge. At this point, then, it may be advisable for the reader to quickly review Chapter 4, "dc and ac Bridge Circuits."

13–5 STRAIN GAGES

All electrical conductors possess some amount of electrical resistance. A bar or wire made of such a conductor will have an electrical resistance that is given by

$$R = \rho\left(\frac{L}{A}\right) \tag{13–1}$$

where R = the resistance in ohms (Ω)
ρ = the **resistivity constant,** a property specific to the conductor material, given in units of ohm-centimeters (Ω-cm)
L = the length in centimeters (cm)
A = the cross-sectional area in square centimeters (cm^2)

■ **Example 13–1**

A constantan (i.e., 55% copper, 45% nickel) round wire is 10 cm long and has a radius of 0.01 mm. Find the electrical resistance in ohms. (*Hint:* The resistivity of constantan is 44.2×10^{-6} Ω-cm).

Solution

$$R = \rho(L/A) \tag{13–1}$$

$$= \frac{(44.2 \times 10^{-6}\ \Omega\text{-cm})(10\ \text{cm})}{\pi\left[0.01\ \text{mm} \times \dfrac{1\ \text{cm}}{10\ \text{mm}}\right]^2}$$

$$= \frac{(4.42 \times 10^{-4}\ \Omega\text{-cm}^2)}{\pi(0.001\ \text{cm})^2}$$

$$= \frac{(4.42 \times 10^{-4}\ \Omega\text{-cm}^2)}{\pi 10^{-6}\ \text{cm}^2} = \mathbf{141\ \Omega}$$

Note that the resistivity factor ρ in Equation (13–1) is a constant, so if length L or area A can be made to vary under the influence of an outside parameter then the electrical resistance of the wire will change. This phenomenon, called **piezoresistivity,** is an example of a transducible property of a material.

Definition: Piezoresistivity is the change in the electrical resistance of a conductor due to changes in length and cross-sectional area. In piezoresistive materials, mechanical deformation of the material produces changes in electrical resistance.

Figure 13–1 shows how an electrical conductor can use the piezoresistivity property to measure **strain,** that is, *forces* applied to it in *compression* or *tension.* Figure 13–1(a) shows a conductor at rest, in which no forces are acting. The length is given as L_0 and the cross-sectional area as A_0. The resistance of this conductor, from Equation (13–1), is

$$R_0 = \rho\left(\frac{L_0}{A_0}\right) \qquad \textbf{(13–2)}$$

where
- ρ = the resistivity as defined previously
- R_0 = the resistance in ohms (Ω) when no forces are applied
- L_0 = the resting (i.e., no force) length in cm
- A_0 = the resting cross-sectional area in cm^2

But Figure 13–1(b) shows the situation where a compression force of magnitude F is applied along the axis in the inward direction. The conductor will deform, causing the length L_1 to decrease to $L_0 - \Delta L$, and the cross-sectional area to increase to $A_0 + \Delta A$. The electrical resistance decreases to $R_0 - \Delta R$:

$$R_1 = (R_0 - \Delta R) \propto \frac{(L_0 - \Delta L)}{(A_0 + \Delta A)} \qquad \textbf{(13–3)}$$

Similarly, when a tension force of the same magnitude (F) is applied (i.e., a force that is directed outward along the axis), the length increases to $L_0 + \Delta L$, and the cross-sectional area decreases to $A_0 - \Delta A$. The resistance will increase to

$$R_2 = (R_0 - \Delta R\ ) \propto \frac{(L_0 + \Delta L)}{(A_0 + \Delta A)} \qquad \textbf{(13–4)}$$

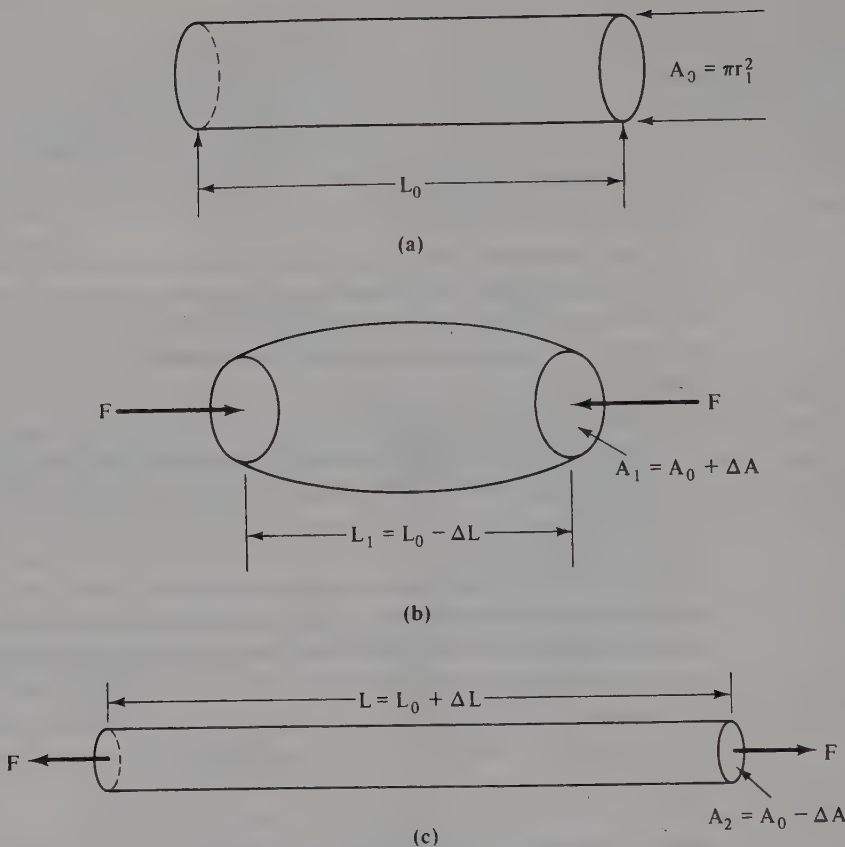


FIGURE 13–1
(a) Unstrained metal bar, (b) metal bar in compression, (c) metal bar in tension.

The **sensitivity** of the strain gage is expressed in terms of unit change of electrical resistance for a unit change in length and is given in the form of a **gage factor** S:

$$S = \frac{(\Delta R/R)}{(\Delta L/L)} \tag{13–5}$$

where
S = the gage factor (dimensionless)
R = the unstrained resistance of the conductor
ΔR = the change in resistance due to strain
L = the unstrained length of the conductor
ΔL = the change in length due to strain

■ **Example 13–2**

Find the gage factor of a 128-Ω conductor that is 24 mm long if the resistance changes 13.3 Ω and the length changes 1.6 mm under a tension force.

Solution

$$S = (\Delta R/R)/(\Delta L/L) \tag{13–5}$$

$$= \frac{(13.3\ \Omega/128\ \Omega)}{(1.6\ \text{mm}/24\ \text{mm})}$$

$$= \frac{1.04 \times 10^{-1}}{6.67 \times 10^{-2}} = \mathbf{1.56}$$

We may also express the gage factor in terms of the length and diameter of the conductor. Recall that the diameter is related to the cross-sectional area ($A = \pi d^2/4 = \pi r^2$), so the relationship between the gage factor S and these other factors is given by

$$S = 1 + 2\left[\frac{(\Delta d/d)}{(\Delta L/L)}\right] \tag{13–6}$$

■ Example 13–3

Calculate the gage factor S if a 1.5-mm-diameter conductor that is 24 mm long changes length by 1 mm and diameter by 0.02 mm under a compression force.

Solution

$$S = 1 + 2\left[\frac{(\Delta d/d)}{(\Delta L/L)}\right] \tag{13–6}$$

$$= 1 + \left[\frac{2[(0.02\ \text{mm})/(1.5\ \text{mm})]}{(1\ \text{mm})/(24\ \text{mm})}\right]$$

$$= 1 + \left[\frac{(2)(1.3 \times 10^{-2})}{(4.2 \times 10^{-2})}\right]$$

$$= 1 + (2)(0.31) = \mathbf{1.62}$$

Note that the expression $\Delta L/L$ is sometimes denoted by the Greek letter ϵ, so Equations (13–5) and (13–6) become

$$S = 1 + \frac{2(\Delta d/d)}{\epsilon}$$

$$S = \frac{(\Delta R/R)}{\epsilon}$$

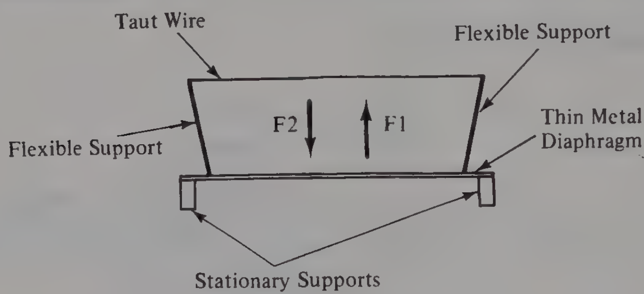
Gage factors for various metals vary considerably. Constantan, for example, has a gage factor of approximately 2, while certain other common alloys have gage factors between 1 and 2. At least one alloy (92% platinum, 8% tungsten) has a gage factor of 4.

Semiconductor materials such as germanium and silicon can be doped with impurities to provide custom gage factors between 50 and 250. The problem with semiconductor strain gages, however, is that they exhibit a marked sensitivity to temperature changes. Where semiconductor strain gages are used, either a thermally controlled environment or temperature compensating circuitry must be provided.

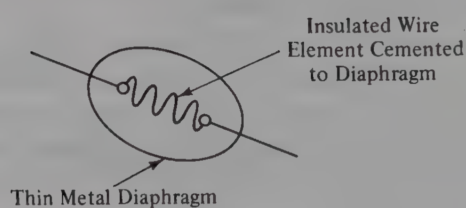
13–6 BONDED AND UNBONDED STRAIN GAGES

Strain gages can be classified as **unbonded** or **bonded.** These categories refer to the method of construction used. Figure 13–2 shows both methods of construction.

The unbonded type of strain gage, shown in Fig. 13–2(a), consists of a wire resistance element stretched taut between two flexible supports. These supports are configured in such a way as to place tension or compression forces on the taut wire when external forces are applied. In the particular example shown, the supports are mounted on a thin metal diaphragm that flexes when a force is applied. Force $F1$ will cause the flexible supports to spread apart, placing a tension force on the wire and increasing its resistance. Alternatively, when force $F2$ is applied, the ends of the flexible supports tend to move closer together, effectively placing a compression force on the wire element, thereby reducing its resistance. In actuality, the wire's resting condition is **tautness,** which



(a)



(b)

FIGURE 13–2
(a) Unbonded strain gage,
(b) bonded strain gage.

implies a tension force. Thus, $F1$ increases the tension force from normal, and $F2$ decreases the normal tension.

The bonded strain gage is shown in Fig. 13–2(b). In this type of device a wire or semiconductor element is cemented to a thin metal diaphragm. When the diaphragm is flexed, the element deforms to produce a resistance change.

The linearity of both types can be quite good, provided that the elastic limits of the diaphragm and the element are not exceeded. It is also necessary to ensure that the ΔL term is only a very small percentage of L.

In the past it has been "standard wisdom"—the opinions of those who make purchase decisions—that bonded strain gages are more rugged, but less linear, than unbonded models. Although this may have been true at one time, recent experience has shown that modern manufacturing techniques produce reliable linear instruments of both types.

13–7 STRAIN GAGE CIRCUITRY

Before a strain gage can be useful, it must be connected into a circuit that will convert its resistance changes to a current or voltage output. Most applications are voltage output circuits.

Figure 13–3(a) shows the **half-bridge** (so called because it is actually half of a Wheatstone bridge circuit) or **voltage divider** circuit. The strain gage element of resistance R is placed in series with a fixed resistance $R1$ across a stable and well-regulated voltage source E. The output voltage E_0 is found from the voltage divider equation:

$$E_0 = \frac{ER}{R + R1} \tag{13–7}$$

Equation (13–7) describes the output voltage E_0 when the transducer is at rest, that is, when nothing is stimulating the strain gage element. When the element is stimulated, however, its resistance changes a small amount ΔR. To simplify our discussion we will adopt the standard convention used in many texts of letting $h = \Delta R$. Thus,

$$E_0 = \frac{E(R + h)}{(R \pm h) + R1} \tag{13–8}$$

Another half-bridge is shown in Fig. 13–3(b), but in this case the strain gage is in series with a **constant current source (CCS),** which will maintain current I at a constant level regardless of changes in strain gage resistance. The normal output voltage E_0 is

$$E_0 = IR \tag{13–9}$$

for nonstimulated conditions, and

$$E_0 = I(R \pm h) \tag{13–10}$$

under stimulated conditions.

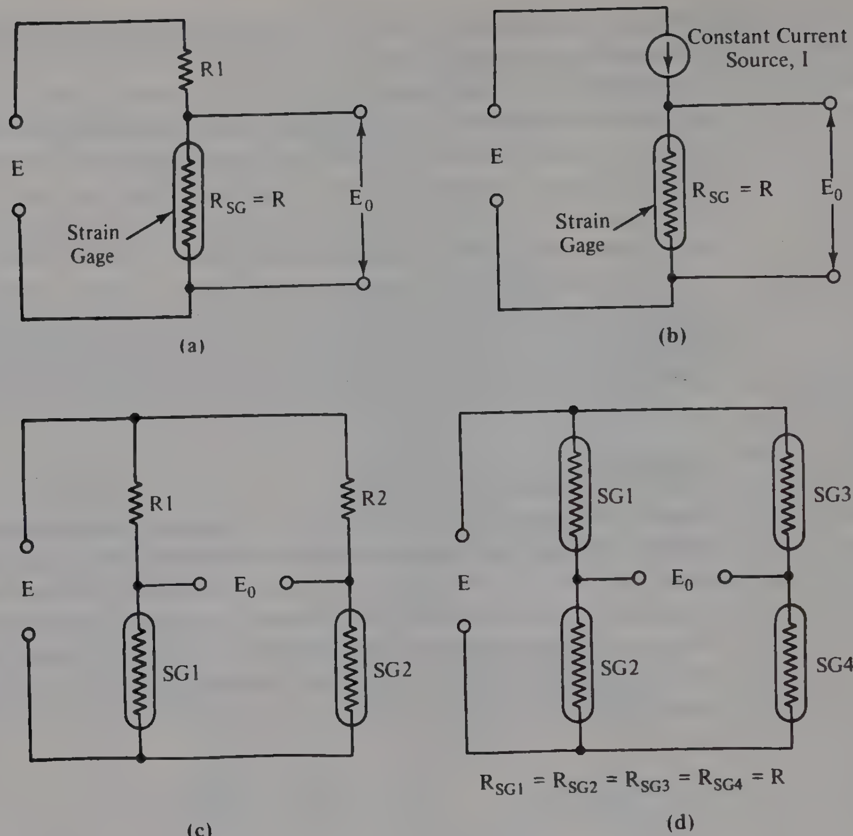


FIGURE 13–3
(a) Constant voltage strain gage circuit (half-bridge type); (b) constant current strain gage circuit (half-bridge type); (c) two strain-gage elements in Wheatstone bridge; (d) four-active-element strain gage Wheatstone bridge.

The half-bridge circuits suffer from one major defect: output voltage E_0 will always be present regardless of the stimulus. Ideally, in any transducer system, we want E_0 to be zero when the stimulus is also zero, and take a value proportional to the stimulus when the stimulus value is nonzero. A Wheatstone bridge circuit in which one or more strain gage elements form the bridge arms has this property.

Figure 13–3(c) shows a circuit in which strain gage elements $SG1$ and $SG2$ form two bridge arms and fixed resistors $R1$ and $R2$ form the other two arms. Usually $SG1$ and $SG2$ will be configured so that their actions oppose each other; that is, under stimulus, $SG1$ will have a resistance $R + h$, and $SG2$ will have a resistance $R - h$, or vice versa.

One of the most linear forms of transducer bridges is the circuit of Fig. 13–3(d) in which all four bridge arms contain strain gage elements. In most such transducers all four strain gage elements have the same resistance, that is, R, which has a value between 100 and 1000 Ω in most cases.

Recall from Chapter 4 that the output voltage from a Wheatstone bridge is the difference between the voltages across the two half-bridge dividers. The following equations hold true for bridges in which one, two, or four equal active elements are used.

One active element:

$$E_0 = \frac{E}{4}\left[\frac{h}{R}\right] \tag{13–11}$$

(accurate to ±5%, provided that $h \leq 0.1$)

Two active elements:

$$E_0 = \frac{E}{2}\left[\frac{h}{R}\right] \tag{13–12}$$

Four active elements:

$$E_0 = \frac{Eh}{R} \tag{13–13}$$

where, for all three equations, E_0 = the output potential in volts (V)
E = the excitation potential in volts (V)
R = the resistance of all bridge arms
h = the quantity ΔR, the change in resistance of a bridge arm under stimulus

(These equations apply only for the case where all the bridge arms have equal resistances under zero stimulus conditions.)

■ Example 13–4

A transducer that measures force has a nominal resting resistance of 300 Ω and is excited by +7.5 V dc. When a 980-dyne force is applied, all four equal-resistance bridge elements change resistance by 5.2 Ω. Find the output voltage E_0.

Solution

$$\begin{aligned} E_0 &= E(h/R) && \text{(13–13)} \\ &= (7.5\ \text{V})(5.2\ \Omega/300\ \Omega) \\ &= (7.5\ \text{V})(5.2)/(300) = \mathbf{0.13\ V} \end{aligned}$$

13–8 TRANSDUCER SENSITIVITY (ψ)

When designing electronic instrumentation systems involving strain gage transducers, it is convenient to use the **sensitivity factor** (denoted by ψ, the Greek letter "psi"), which relates the output voltage in terms of the excitation voltage and the applied stimulus. In most cases we see a specification giving the number of microvolts or millivolts output per volt of excitation potential per unit of applied stimulus (i.e., $\mu V/V/Q_0$ or $mV/V/Q_0$). Thus,

$$\psi = E'_0/V/Q_0 \tag{13–14a}$$

and

$$\psi = \frac{E'_0}{V \times Q_0} \tag{13–14b}$$

where E'_0 = the output potential
V = one unit of potential, that is, 1 V
Q_0 = one unit of stimulus

The sensitivity is often given as a specification by the transducer manufacturer. From it we can predict output voltage for any level of stimulus and excitation potential. The output voltage, then, is found from

$$E_0 = \psi EQ \tag{13–15}$$

where E_0 = the output potential in volts (V)
ψ = the sensitivity in $\mu V/V/Q_0$
E = the excitation potential in volts (V)
Q = the stimulus parameter

■ **Example 13–5**

A well-known medical arterial blood pressure transducer uses a four-element piezoresistive Wheatstone bridge with a sensitivity of 5 µV per volt of excitation per torr of pressure, that is, 5 µV/V/T (*Note:* 1 torr = 1 mmHg). Find the output voltage if the bridge is excited by 5 V dc and 120 T of pressure is applied.

Solution

$$E_0 = \psi EQ \tag{13–15}$$

$$= \frac{5\ \mu\text{V}}{\text{V}-\text{T}} \times (5\ \text{V}) \times (120\ \text{T})$$

$$= (5 \times 5 \times 120)\ \mu\text{V} = \mathbf{3000\ \mu V}$$

13–9 BALANCING AND CALIBRATING THE BRIDGE

Few, if any, Wheatstone bridge strain gages meet the ideal condition in which all four arms have exactly equal resistances. In fact, the bridge resistance specified by the manufacturer is a *nominal* value only. There will inevitably be an **offset voltage,** that is, $E_0 \neq 0$ when $Q = 0$. Figure 13–4 shows a circuit that will balance the bridge when the stimulus is zero. Potentiometer $R1$, usually a type with 10 or more turns of operation, is used to inject a balancing current I into the bridge circuit at one of the nodes. $R1$ is adjusted, with the stimulus at zero, for zero output voltage.

The best calibration method is to apply a precisely known value of stimulus to the transducer and adjust the amplifier following the transducer for the output proper for that level of stimulus. But that may prove unreasonably difficult in some cases, so an *artificial* calibrator is needed to simulate the stimulus. This function is provided by $R3$ and $S1$ in Fig. 13–4. When $S1$ is open, the transducer is able to operate normally, but when $S1$ is closed, it *unbalances* the bridge and produces an output voltage E_0 that simulates some standard value of the stimulus. The value of $R3$ is given by

$$R3 = \left[\frac{R}{4Q\psi} - \frac{R}{2}\right] \qquad \textbf{(13–16)}$$

where $R3$ = the resistance of R3 in ohms (Ω)
R = the nominal resistance of the bridge arms in ohms (Ω)
Q = the calibrated stimulus parameter
ψ = the sensitivity factor in $V/V/Q$ (*Note:* Different units of ψ are used: V instead of μV.)

■ Example 13–6

An arterial blood pressure transducer has a sensitivity of 10 μV/V/T and a nominal bridge arm resistance of 200 Ω. Find a value for $R3$ in Fig. 13–4 if we want the calibration to simulate an arterial pressure of 200 mmHg (i.e., 200 T).

Solution

$$R3 = \left[\frac{R}{4Q\psi} - \frac{R}{2}\right] \qquad \textbf{(13–16)}$$

$$= \frac{200\ \Omega}{4 \times \left(\dfrac{10^{-5}\ \text{V}}{\text{V}-\text{T}}\right) \times 200} - \frac{200\ \Omega}{2}$$

$$= \frac{200\ \Omega}{(4)(10^{-5})(200)} - 100\ \Omega = \mathbf{24{,}900\ \Omega}$$

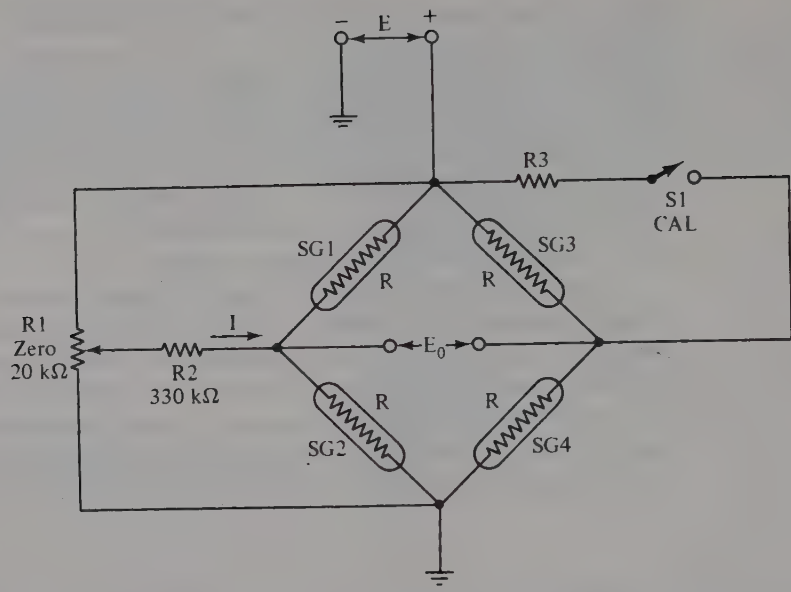


FIGURE 13–4
Circuit for using Wheatstone bridge transducer.

13–10 TEMPERATURE TRANSDUCERS

A large number of physical phenomena are temperature dependent, so we find quite a variety of electrical temperature transducers on the market. In this discussion, however, we will discuss only three basic types: **thermistor, thermocouple,** and **semiconductor pn junctions.**

13–11 THERMISTORS

Metals and most other conductors are temperature sensitive and will change electrical resistance with changes in temperature, namely,

$$R_t = R_0[1 + \alpha(T - T_0)] \tag{13–17}$$

where R_t = the resistance in ohms (Ω) at temperature T
R_0 = the resistance in ohms (Ω) at temperature T_0 (often a standard reference temperature)
T = the temperature of the conductor
T_0 = a previous temperature of the conductor at which R_0 was determined
α = the **temperature coefficient** of the material, a property of the conductor, in °C^{-1}

The temperature coefficients of most metals are positive, as are the coefficients for most semiconductors; for example, gold has a value of +0.004/°C. Ceramic semiconductors used to make **thermistors** (i.e., thermal resistors) can have either negative or positive temperature coefficients depending upon their composition.

The resistance of a thermistor is given by

$$R_t = R_0 e^{\beta[(1/T) - 1/T_0)]}$$

where R_t = the resistance of the thermistor at temperature T
R_0 = the resistance of the thermistor at a reference temperature (usually the ice point, 0°C, or room temperature, 25°C)
e = the base of the natural logarithms
T = the thermistor temperature in kelvin (K)
T_0 = the reference temperature in kelvin (K)
β = a property of the material used to make the thermistor (*Note:* β will usually have a value between 1500 K and 7000 K.)

■ Example 13–7

Calculate the resistance of a thermistor at 100°C if the resistance at 0°C is 18 kΩ. The material of the thermistor has a value of 2200 K. (*Note:* 0°C = 273 K, so 100°C = 373 K.)

Solution

$$R_t = R_0 \exp \beta \left(\frac{1}{T} - \frac{1}{T_0} \right) \tag{13–18}$$

$$= (1.8 \times 10^4 \ \Omega) \exp\left[(2200 \text{ K}) \left(\frac{1}{373 \text{ K}} - \frac{1}{273 \text{ K}} \right) \right]$$

$$= 1.8 \times 10^4 \ \Omega \ e^{-2.15} = \mathbf{2084 \ \Omega}$$

Equation (13–18) demonstrates that the response of a thermistor is exponential, as shown in Fig. 13–5. Note that both curves are nearly linear over a portion of their ranges, but then become decidedly nonlinear in the remainder of the region. If a wide measurement range is needed, then a linearization network will be required.

Thermistor transducers will be used in any of the circuits in Fig. 13–3. They will also be found using many packaging arrangements. Figure 13–6 shows a bead thermistor used in medical instruments to continuously monitor a patient's rectal temperature.

The equations governing thermistors usually apply if there is little **self-heating** of the thermistor, although there are applications where self-heating is used. But in straight temperature measurements it is to be avoided. To minimize self-heating it is necessary to control the power dissipation of the thermistor.

Also of concern in some applications is the **time constant** of the thermistor. The resistance does not jump immediately to the new value when the temperature changes but requires a small amount of time to stabilize at the new resistance value. This is expressed

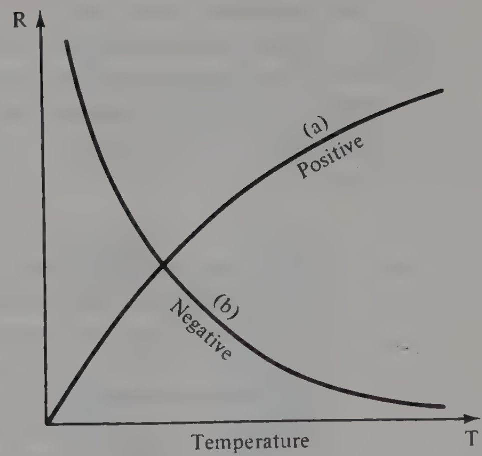


FIGURE 13–5
Thermistor temperature-vs.-resistance curves.

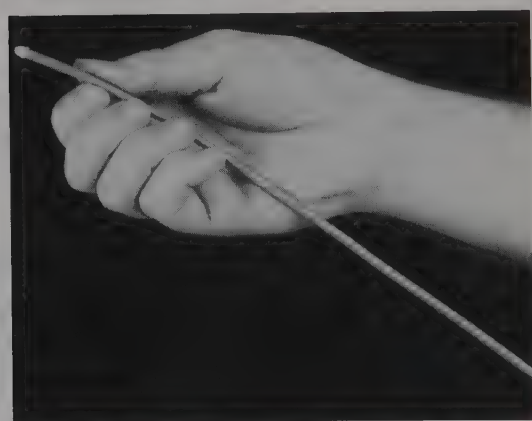


FIGURE 13–6
Thermistor in a medical rectal probe (courtesy of Electronics-for-Medicine).

in terms of the time constant of the thermistor in a manner that is reminiscent of capacitors charging in RC circuits.

13–12 THERMOCOUPLES

When two dissimilar metals are joined together to form a "vee" [as in Fig. 13–7(a)], it is possible to generate an electrical potential merely by heating the junction. This phenomenon, first noted by Seebeck in 1823, is due to different *work functions* for the two metals. Such a junction is called a **thermocouple,** The Seebeck EMF generated by the junction is proportional to the junction temperature and is reasonably linear over wide temperature ranges.

A simple thermocouple, shown in Fig. 13–7(b), uses two junctions. One junction is the measurement junction and is used as the thermometry probe. The other junction is a reference and is kept at a reference temperature such as the ice point (0°C) or room temperature.

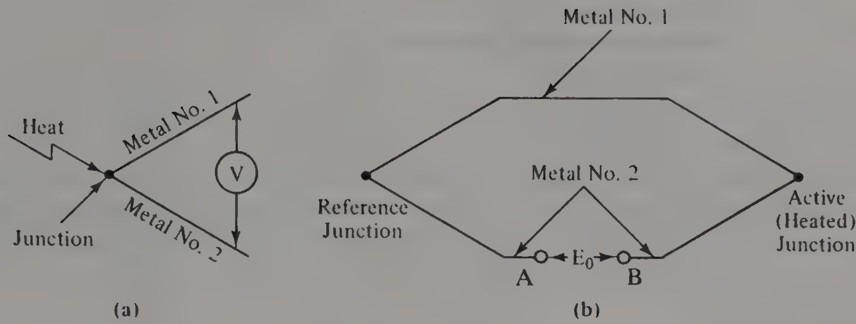


FIGURE 13–7
(a) Thermocouple junction, (b) two-thermocouple temperature transducer.

Interestingly enough, there is an inverse thermocouple phenomenon, called the **Peltier effect,** in which an electrical potential applied across A–B in Fig. 13–7(b) will cause one junction to absorb heat (i.e., get hot) and the other to lose heat (i.e., get cold). Semiconductor thermocouples have been used in small-scale environmental temperature chambers, (e.g., drink coolers), and it is reported that one company researched the possibility of using Peltier devices to cool submarine equipment. Ordinary air conditioning equipment proves too noisy in submarines desirous of "silent running."

13–13 SEMICONDUCTOR TEMPERATURE TRANSDUCERS

Ordinary pn junction diodes exhibit a strong dependence upon temperature. This effect can be easily demonstrated by using an ohmmeter and an ordinary rectifier diode such as the 1N4000-series devices. Connect the ohmmeter so that it forward-biases the diode, and note the resistance at room temperature. Next hold a soldering iron or other heat source close to the diode's body, and watch the electrical resistance change. In a circuit such as Fig. 13–8 the current is held constant, so output voltage E_0 will change with temperature-caused changes in diode resistance.

Another solid-state temperature transducer is shown in Fig. 13–9. In this version the temperature-sensor device is a pair of diode-connected transistors. In any transistor the base-emitter voltage V_{be} is

$$V_{be} = \frac{kT}{q} \ln\left[\frac{I_c}{I_s}\right] \tag{13–19}$$

where V_{be} = the base-emitter potential in volts (V)
k = Boltzmann's constant (1.38×10^{-23} J/K)
T = the temperature in kelvin (K)
Q = the electronic charge (1.6×10^{-19} C)
ln denotes the natural logarithms
I_c = the collector current in amperes (A)
I_s = the reverse saturation current in amperes (A)

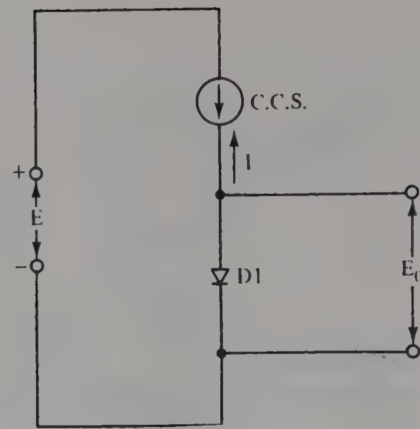


FIGURE 13–8
pn junction diode as a temperature transducer.

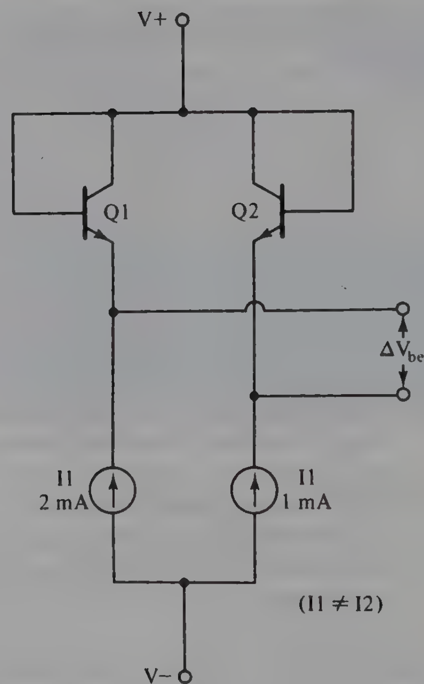


FIGURE 13–9
Two transistors connected as diodes from a temperature transducer if $I1 = I2/2$.

Note that the k and Q terms in Equation (13–19) are constants, and both currents can be made constant. The only variable, then, is *temperature.*

The circuit of Fig. 13–9 uses two transistors connected to provide a differential output voltage ΔV_{be} that is the difference between $V_{be(Q1)}$ and $V_{be(Q2)}$. Combining the expressions for V_{be} for both transistors yields the expression

$$\Delta V_{be} = \frac{kT}{Q} \ln\left[\frac{I1}{I2}\right] \tag{13–20}$$

Note that since ln 1 = 0, currents $I1$ and $I2$ *must not be equal.* In general, designers set a ratio of 2 : 1; that is, $I1 = 2$ mA and $I2 = 1$ mA. Since currents $I1$ and $I2$ are supplied from constant current sources, the ratio $I1/I2$ is a constant. Also, the logarithm of a constant is a constant. Therefore, all terms in Equation (13–20) are constants except temperature T. Equation (13–20), therefore, may be written in the form

$$\Delta V_{be} = KT \tag{13–21}$$

where $K = (k/Q)\ \ln(I1/I2)$
$K = (1.38 \times 10^{-23})/(1.6 \times 10^{-19})\ \ln(2/1)$
$K = 5.98 \times 10^{-5}$ V/K = 59.8 μV/K

We may now rewrite Equation (13–21) in the following form:

$$\Delta V_{be} = 59.8\ \mu\text{V/K} \tag{13–22}$$

■ **Example 13–8**

Calculate the output voltage from a circuit such as Fig. 13–9 if the temperature is 35°C. (*Hint:* K = °C + 273.)

Solution

$$\Delta V_{be} = KT \tag{13–21}$$

$$= \frac{59.8\ \mu\text{V}}{\text{K}} \times (35 + 273)\ \text{K}$$

$$= (59.8)(308)\ \mu\text{V} = 18{,}418\ \mu\text{V} = \mathbf{0.0184\ V}$$

In most thermometers using the circuit of Fig. 13–9, an amplifier increases the output voltage to a level that is numerically the same as a unit of temperature, so that the temperature may be easily read from a digital voltmeter. The most common scale factor is 10 mV/K, so for our transducer the postamplifier requires a gain of

$$A_v = \frac{10\ \text{mV/K}}{59.8\ \mu\text{V} \times \dfrac{1\ \text{mV}}{10^3\ \mu\text{V}}} = 167$$

13–14 INDUCTIVE TRANSDUCERS

Inductance L and inductive reactance X_L are transducible properties because they can be *varied* by certain mechanical methods.

Figure 13–10(a) shows an example of an inductive Wheatstone bridge. Resistors $R1$ and $R2$ form two fixed arms of the bridge, while coils $L1$ and $L2$ form variable arms. Since inductors are used, the excitation voltage must be ac. In most cases the ac excitation source will have a frequency between 400 and 5000 Hz and an rms amplitude of 5 to 10 V.

The inductors are constructed coaxially, as shown in Fig. 13–10(b), with a common core. It is a fundamental property of any inductor that a ferrous core increases its inductance. In the rest condition (i.e., zero-stimulus) the core will be positioned equally inside both coils. If the stimulus moves the core in the direction shown in Fig. 13–10(b), the core tends to move out of $L1$ and further into $L2$. This action reduces the inductive reactance of $L1$ and increases that of $L2$, unbalancing the bridge.

13–15 LINEAR VARIABLE DIFFERENTIAL TRANSFORMERS (LVDTs)

Another form of inductive transformer is the **linear variable differential transformer (LVDT)** shown in Fig. 13–11. The construction of the LVDT is similar to that of the inductive bridge, except that it also contains a primary winding.

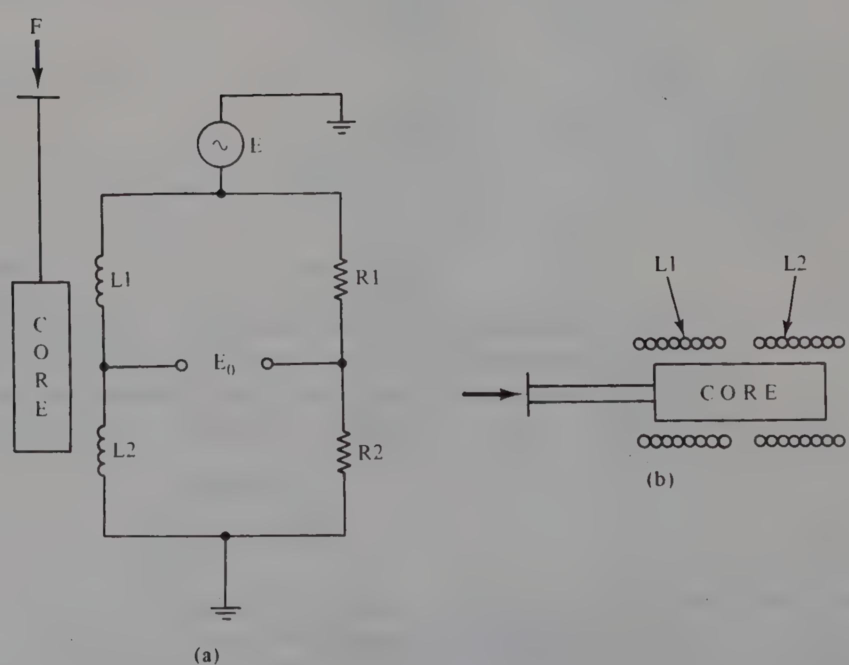


FIGURE 13–10
(a) Inductive Wheatstone bridge transducer, (b) mechanical form.

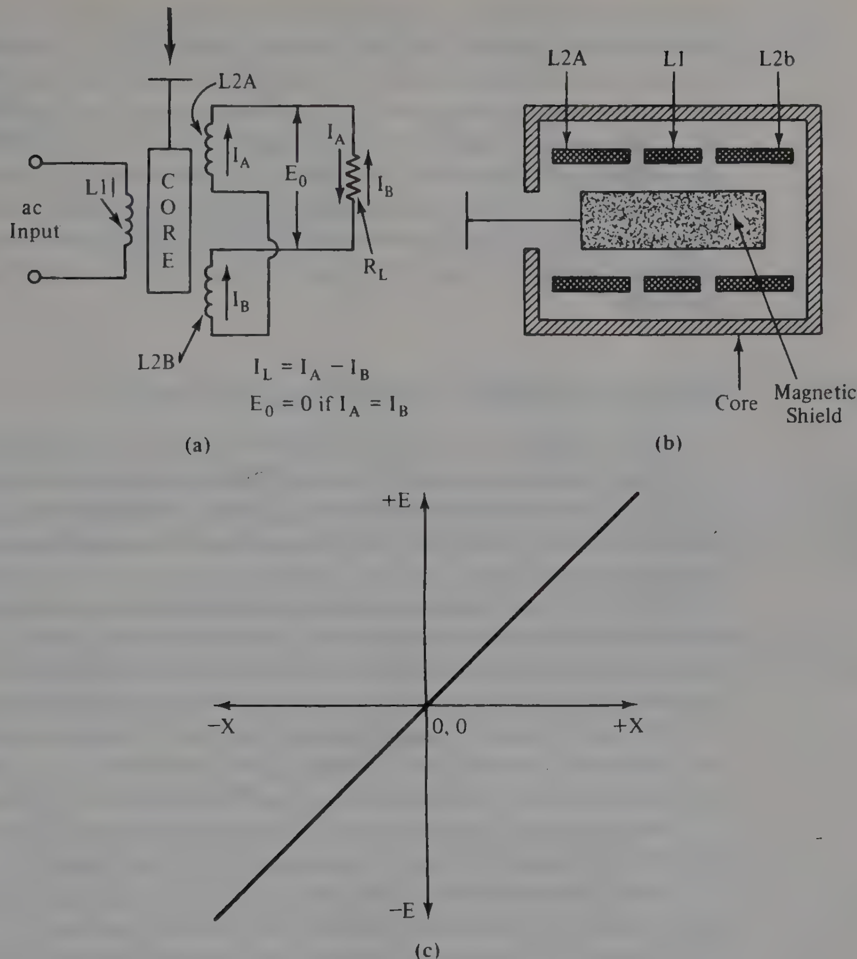


FIGURE 13–11
(a) LVDT, (b) LVDT construction, (c) output transfer function.

One advantage of the LVDT over the bridge-type transducer is that it provides higher output voltages for small changes in core position. Several commercial models are available that produce 50 to 300 mV/mm. In the latter case a 1-mm displacement of the core produces a voltage output of 300 mV.

In normal operation the core is equally inside both secondary coils, *L2a* and *L2b*, and an ac carrier is applied to the primary winding. This carrier typically has a frequency between 40 Hz and 20 kHz and an amplitude in the range 1 to 10 V rms.

Under rest conditions the coupling between the primary and each secondary is equal. The currents flowing in each secondary, then, are equal to each other. Note in Fig. 13–11(a) that the secondary windings are connected in series opposing, so if the secondary winding currents are equal, they will exactly cancel each other in the load. The ac voltage appearing across the load, therefore, is *zero* ($I_a = I_b$).

But when the core is moved so that it is more inside *L2b* and less inside *L2a*, the coupling between the primary and *L2b* is greater than the coupling between the primary and *L2a*. Since this fact makes the two secondary currents no longer equal, the cancellation is not complete. The current in the load I_L is no longer zero. The output voltage appearing across load resistor R_L is proportional to the core displacement, as shown in Fig. 13–11(c). The *magnitude* of the output voltage is proportional to the *amount* of core displacement, while the *phase* of the output voltage is determined by the *direction* of the displacement.

13–16 POSITION-DISPLACEMENT TRANSDUCERS

A position transducer will create an output signal that is proportional to the position of some object along a given axis. For very small position ranges we could use a strain gage (i.e., Fig. 13–12), but note that the range of such transducers is necessarily very small. Most strain gages either are nonlinear for large displacements or are damaged by large displacements.

The LVDT can be used as a position transducer. Recall that the output polarity indicates the direction of movement from a zero-reference position, and the amplitude indicates the magnitude of the displacement. Although the LVDT will accommodate larger displacements than the strain gage, it is still limited in range.

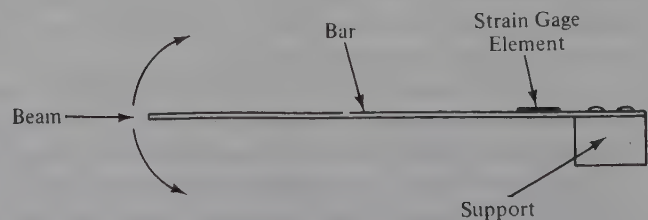
The most common form of position transducer is the potentiometer. For applications that are not too critical, ordinary linear taper potentiometers are often sufficient. Rotary models are used for curvilinear motion, and slide models for rectilinear motion.

We have already seen an example of the potentiometer as position transducers in our discussion (Chapter 10) on servomechanism strip chart recorders.

In precision applications designers use either regular precision potentiometers or special potentiometers designed specifically as position transducers.

Figure 13–13 shows two possible circuits using potentiometers as position transducers. In Fig. 13–13(a) we see a single-quadrant circuit for use where the zero point (i.e., the starting reference) is at one end of the scale. The pointer will always be at some point such that $0 \le x \le X_m$. The potentiometer is connected so that one end is grounded and the other is connected to a precision, regulated voltage source $V+$. The value of V_x represents X and will be $0 \le V_x \le V+$, such that $V_x = 0$ when $X = 0$, and $V_x = V+$ when $X = X_m$.

FIGURE 13–12
Beam transducer using a strain gage element.



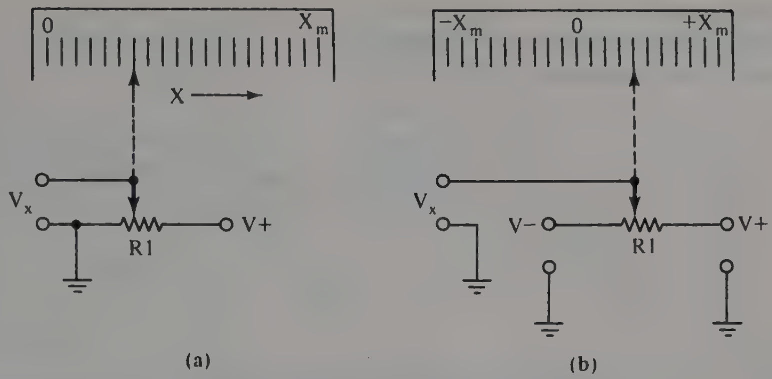


FIGURE 13–13
(a) Position transducer using a potentiometer (one quadrant), (b) position transducer using a potentiometer (two quadrants).

A two-quadrant system, shown in Fig. 13–13(b), is similar to the previous circuit except that instead of grounding one end of the potentiometer, it is connected to a precision, regulated, *negative-to-ground* power source, *V*–. Figure 13–14 shows the output functions of these two transducers. Figure 13–14(a) represents the circuit of Fig. 13–13(a), while Figure 13–14(b) represents the circuit of Fig. 13–13(b).

A four-quadrant transducer can be made by placing two circuits such as Fig. 13–13(b) at right angles to each other and arranging linkage so that the output signal varies appropriately.

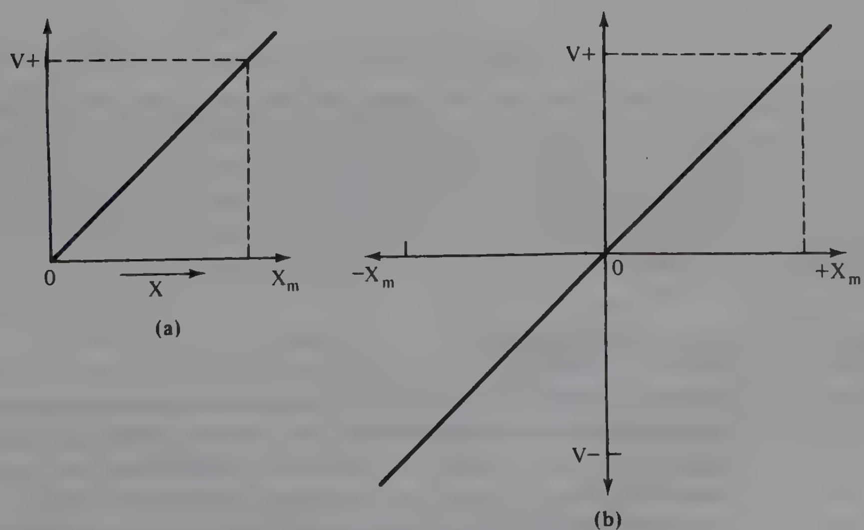
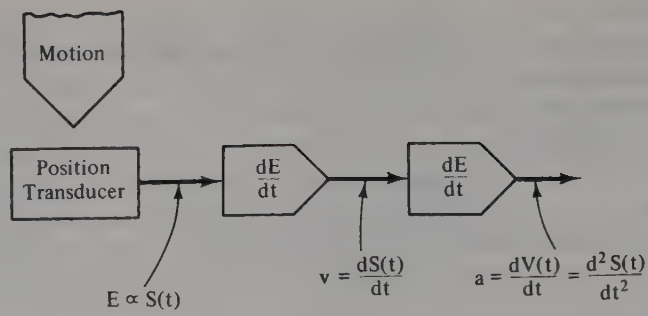


FIGURE 13–14
(a) *V*+ vs. *X* for Fig. 13–13(a); (b) *V* vs. *X* for Fig. 13–13(b).

FIGURE 13–15
Example of derived signals using integration or differentiation.



13–17 VELOCITY AND ACCELERATION TRANSDUCERS

Velocity can be defined as displacement per unit of time, and **acceleration** is the time rate of change of velocity. Since both velocity v and acceleration a can be related to position s, position transducers are often used to *derive* velocity and acceleration signals. The relationships are as follows:

$$v = \frac{ds}{dt} \tag{13–23}$$

$$a = \frac{dv}{dt} \tag{13–24}$$

$$a = \frac{d^2s}{dt^2} \tag{13–25}$$

Velocity and acceleration are the first and second time derivatives of displacement (i.e., change of position), respectively. We may derive electrical signals proportional to v and a by using an operational amplifier differentiator circuit (see Fig. 13–15). The output of the transducer is a time-dependent function of position (i.e., displacement). This signal is differentiated by the stages following to produce the velocity and acceleration signals.

13–18 TACHOMETERS

Alternating current and direct current generators are also used as velocity transducers. In their basic form they will transduce rotary motion—that is, produce an angular velocity signal—but with appropriate mechanical linkage they will also indicate rectilinear motion.

In the case of a dc generator, the output signal is a dc voltage with a magnitude that is proportional to the angular velocity of the armature shaft.

The ac generator, or **alternator,** maintains a relatively constant output voltage, but its ac *frequency* is proportional to the angular velocity of the armature shaft.

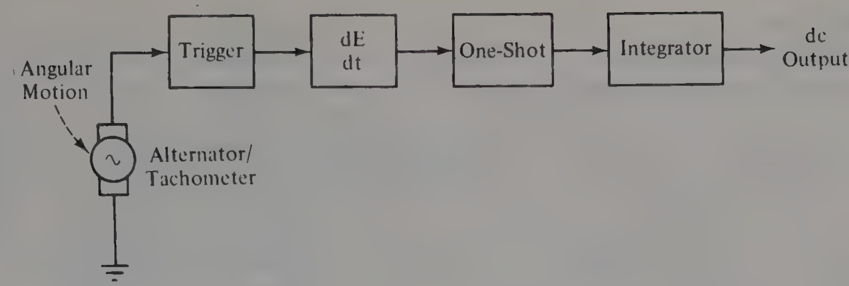


FIGURE 13–16
Alternator tachometer.

If a dc output is desired instead of an ac signal, then a circuit similar to Fig. 13–16 is used. The ac output of the tachometer is fed to a trigger circuit (either a comparator or a Schmitt trigger) so that squared-off pulses are created. These pulses are then differentiated to produce spike-like pulses to trigger the monostable multivibrator (one-shot). The output of the one-shot is integrated to produce a dc level proportional to the tachometer frequency.

The reason for using the one-shot stage is to produce output pulses that have a *constant amplitude* and *duration.* Only the pulse repetition rate (i.e., the number of pulses per unit of time) varies with the input frequency. This fact allows us to integrate the one-shot output to obtain our needed dc signal. If either duration or amplitude varied, then the integrator output would be meaningless. This technique, incidentally, is widespread in electronic instruments, so you should understand it well.

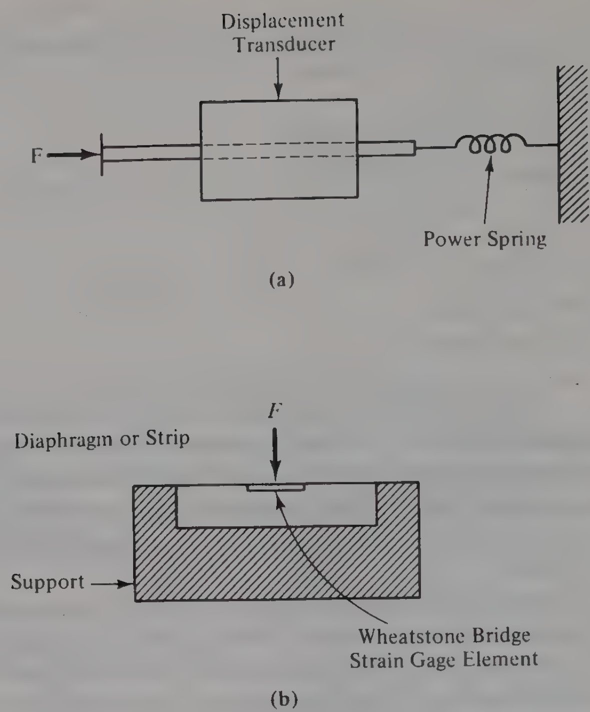
13–19 FORCE AND PRESSURE TRANSDUCERS

Force transducers can be made by using strain gages or either LVDT or potentiometer displacement transducers. The displacement transducer [Fig. 13–17(a)] becomes a force transducer by causing a power spring to either compress or stretch. Recall Hooke's law, which tells us that the force required to compress or stretch a spring is proportional to a *constant* and the *displacement* caused by the compression or tension force applied to the spring. So by using a displacement transducer and a calibrated spring, we are able to measure force.

Strain gages connected to flexible metal bars (refer to Fig. 13–12) are also used to measure force, because a certain amount of force is required to deflect the bar any given amount. Several transducers on the market use this technique, and they are advertised as "force-displacement" transducers. Such transducers form the basis of the digital bathroom scales now on the market.

Do not be surprised to see such transducers, especially the smaller types, calibrated in *grams.* We all know that the gram is a unit of *mass,* not force, so what this usage refers to is the gravitational force on one gram (1 g) at the earth's surface, roughly 980 dynes. A 1-g weight suspended from the end of the bar in Fig. 13–12 would represent a force of 980 dynes.

FIGURE 13–17
(a) Force from a displacement transducer, (b) force/pressure transducer.



A side view of a cantilever force transducer is shown in Fig. 13–17(b). In this device a flexible strip is supported by mounts at either end, and a piezoresistive strain gage is mounted to the underside of the strip. Flexing the strip unbalances the gage's Wheatstone bridge, producing an output voltage.

A related device uses a cup- or barrel-shaped support and a circular diaphragm instead of the strip. Such a device will measure force or pressure, that is, *force per unit of area.*

13–20 FLUID PRESSURE TRANSDUCERS

Fluid pressures are measured in a variety of ways, but the most common involve a transducer such as those shown in Figs. 13–18 and 13–19.

In the example of Fig. 13–18(a), a strain gage or LVDT is mounted inside a housing that has a bellows or an aneroid assembly exposed to the fluid. More force is applied to the LVDT or gage assembly as the bellows compresses. The compression of the bellows is proportional to the fluid pressure.

An example of the **Bourdon tube** pressure transducer is shown in Fig. 13–18(b). Such a tube is hollow and curved, but flexible. When a pressure is applied through the inlet port, the tube tends to straighten out. If the end tip is connected to a position/displacement transducer, then the transducer output will be proportional to the applied pressure.

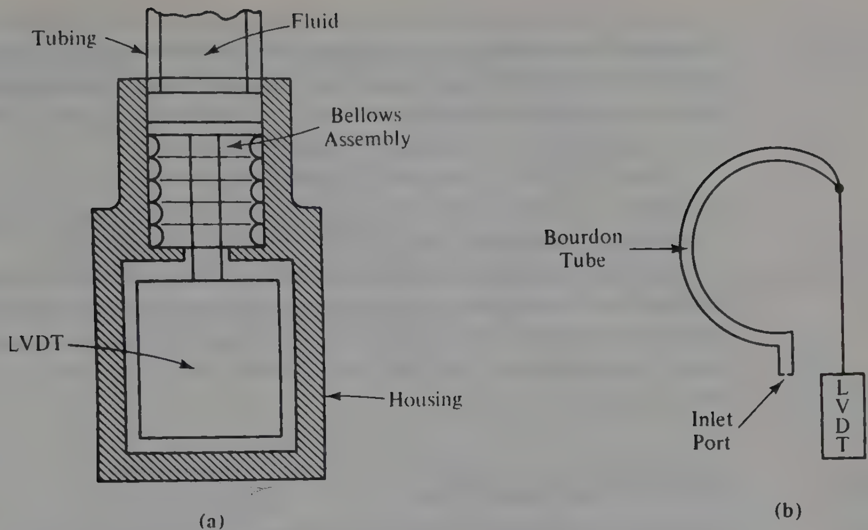


FIGURE 13–18
(a) Fluid pressure transducer, (b) Bourdon tube fluid pressure transducer.

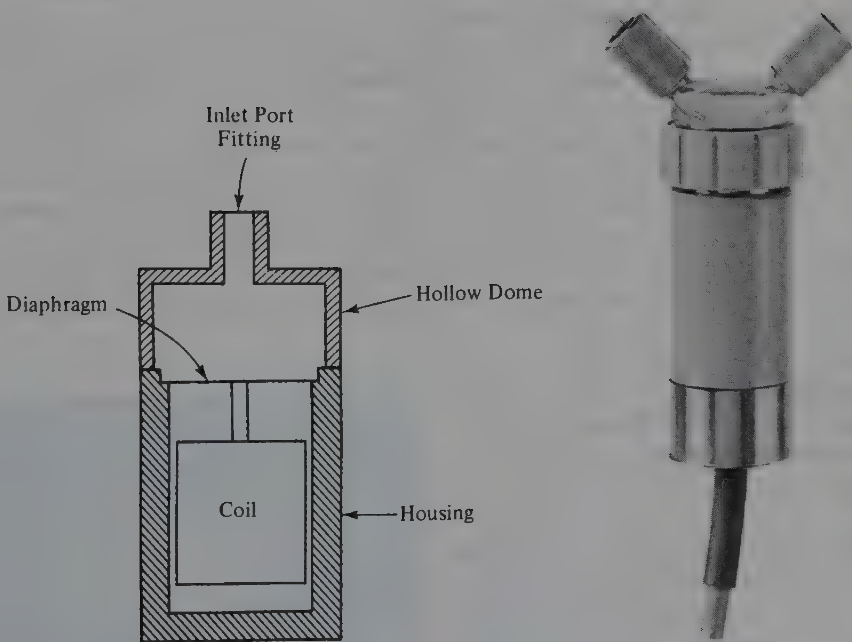


FIGURE 13–19
(a) Dome-type fluid pressure transducer, (b) commercial dome-type transducer [(b) courtesy of Hewlett-Packard].

Figure 13–19(a) shows another popular form of fluid transducer. In this version a diaphragm is mounted on a cylindrical support similar to that in Fig. 13–18. In some cases a bonded strain gage is attached to the underside of the diaphragm, or flexible supports to an unbonded type are used. In the example shown, the diaphragm is connected to the core drive bar of an inductive transducer or LVDT.

Figure 13–19(b) shows the Hewlett-Packard type-1280 transducer used in medical electronics to measure human blood pressure. In this device the hollow fluid-filled dome is fitted with **Luer-lock** fittings, standards in medical apparatus.

The fluid transducers shown so far will measure gage pressure (i.e., pressure above atmospheric pressure) because one side of the diaphragm is open to air. A **differential** pressure transducer will measure the difference between pressures applied to the two sides of the diaphragm. Such devices will have two ports marked "*P*1" and "*P*2" or something similar.

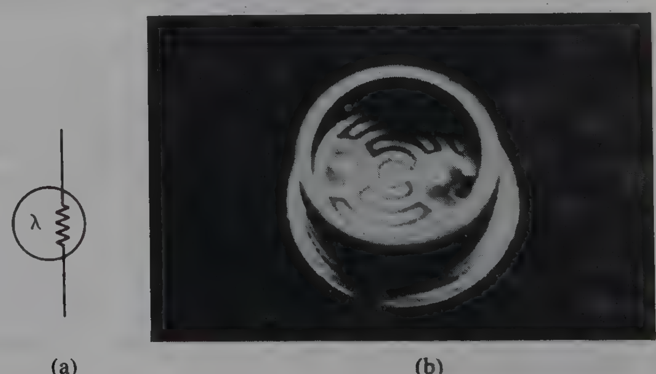
13–21 LIGHT TRANSDUCERS

There are several different phenomena for measuring light, and they create different types of transducers. For this chapter we will limit the discussion to **photoresistors, photovoltaic cells, photodiodes,** and **phototransistors.**

A photoresistor can be made because certain semiconductor elements show a marked decrease in electrical resistance when exposed to light. Most materials do not change linearly with increased light intensity, but certain combinations such as cadmium sulfide (CdS) and cadmium selenide (CdSe) are effective. These cells operate over a spectrum from "near-infrared" through most of the visible light range, and they can be made to operate at light levels of 10^{-3} to 10^{+3} footcandles (i.e., 10^{-3} to 70 mW/cm^2). Figure 13–20(a) shows the photoresistor circuit symbol, and Figure 13–20(b) shows an example of a photoresistor.

A photovoltaic cell, or "solar cell," as it is sometimes called, will produce an electrical current when connected to a load. Both silicon (Si) and selenium (Se) types are known. The Si type covers the visible and near-infrared spectrum, at intensities between 10^{-3} and 10^{+3} mW/cm^2. The selenium cell, on the other hand, operates at intensities of 10^{-1} to 10^2 mW/cm^2, but it accepts a spectrum of near-infrared to the ultraviolet.

FIGURE 13–20
(a) Symbol for photoresistor cell,
(b) actual photoresistor cell.



Semiconductor pn junctions under sufficient illumination will respond to light. Interestingly enough, they tend to be photoconductive when heavily reverse biased, and photovoltaic when forward biased. These phenomena have led to a whole family of photodiodes and phototransistors.

13–22 CAPACITIVE TRANSDUCERS

A parallel plate capacitor can be made by positioning two conductive planes parallel to each other. The capacitance is given by

$$C = \frac{kKA}{d} \tag{13–26}$$

where C = the capacitance in *farads* (F) or a subunit (µF, pF, etc.)
k = a units constant
K = the dielectric constant of the material used in the space between the plates (K for air is 1)
A = the area of the plates "shading" each other
d = the distance between the plates

Figure 13–21 shows several forms of capacitance transducers. In Fig. 13–21(a) we see a rotary plate capacitor that is not unlike the variable capacitors used to tune radio transmitters and receivers. The capacitance of this unit is proportional to the amount of area on the fixed plate that is covered (i.e., "shaded") by the moving plate. This type of transducer will give signals proportional to curvilinear displacement or angular velocity.

A rectilinear capacitance transducer, shown in Fig. 13–21(b), consists of a fixed cylinder and a moving cylinder. These pieces are configured so that the moving piece fits inside the fixed piece but is insulated from it.

Two types of capacitive transducers discussed so far vary capacitance by changing the shaded area of two conductive surfaces. Figure 13–21(c), on the other hand, shows a transducer that varies the *spacing* between surfaces, that is, the d term in Equation (13–26). In this device the metal surfaces are a fixed plate and a thin diaphragm. The dielectric is either air or a vacuum. Such devices are often used as capacitance microphones.

Capacitance transducers can be used in several ways. One method is to use the varying capacitance to frequency-modulate an RF oscillator. This method is employed with capacitance microphones [those built as in Fig. 13–21(c), not the electrotet type]. Another method is to use the capacitance transducer in an ac bridge circuit.

Figure 13–22 shows an application in which a differential capacitance transducer is used to provide a position signal in a PMMC galvanometer chart recorder (Chapter 10). This type of transducer is superior to others in this application because it can be constructed with very low mass, so it will not significantly dampen the pen motion. The position signal is used in a feedback control circuit that gives the pen controlled damping. In one model the differentiated position signal warns the instrument that the pen will slam into the mechanical limit stop, and it puts the brakes on to prevent pen damage.

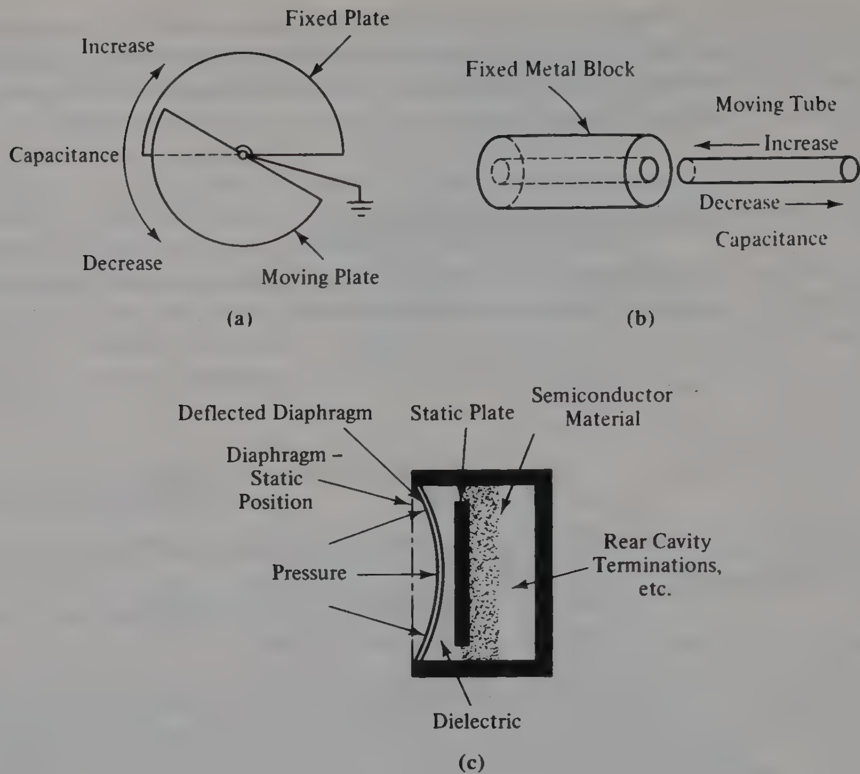


FIGURE 13–21
Capacitance transducers (from Harry Thomas, *Handbook of Biomedical Instrumentation and Measurement,* Figure 1–6, p. 13, Reston Publishing Co.).

SUMMARY

1. A **transducer** is a device that converts a physical stimulus such as displacement, force, temperature, and so on, to an electrical voltage or current proportional to the magnitude of the stimulus.
2. A **strain gage** depends upon the **piezoresistance**—that is, resistance change due to mechanical deformation—phenomenon.
3. There are two basic forms of inductive transducers: X_L bridge and LVDT.
4. Capacitive transducers depend upon varying the space between, or the shaded area of, two conductive surfaces.
5. Several types of temperature transducers are common: thermistor, thermocouple, and semiconductor junctions.

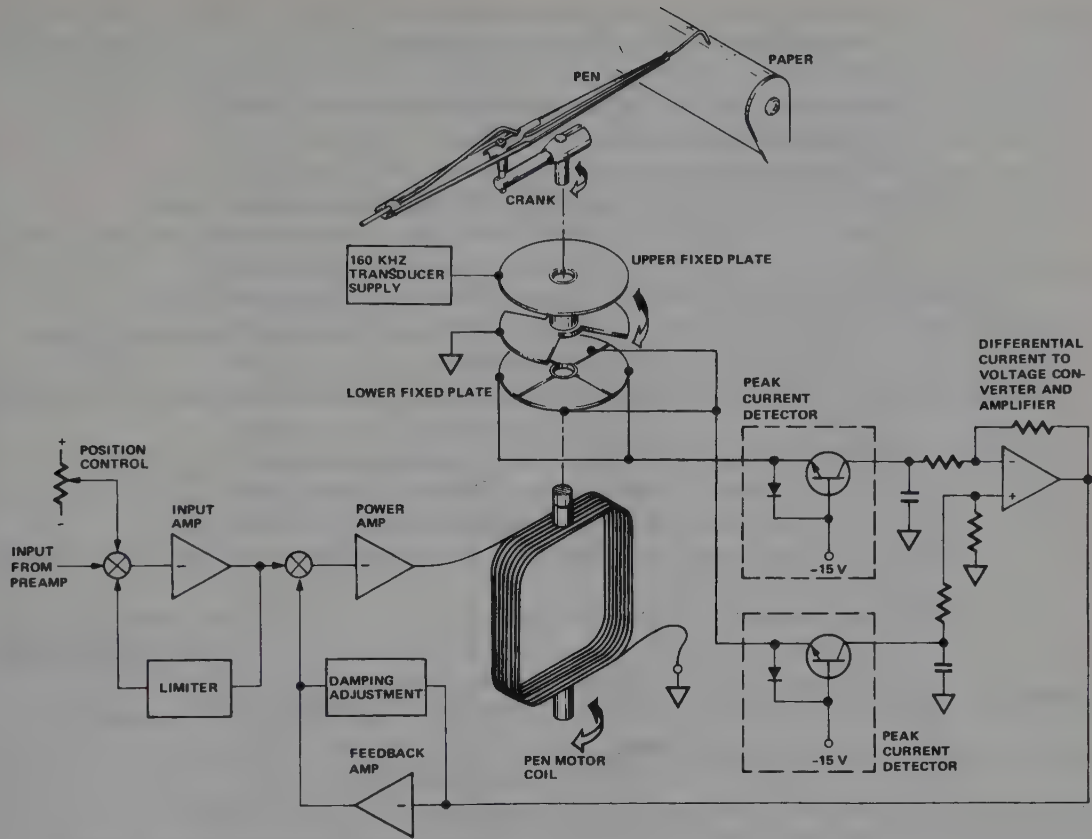


FIGURE 13–22
Capacitance transducers used as a position/velocity sensor (courtesy of Hewlett-Packard).

RECAPITULATION

Now go back and try to answer the questions at the beginning of the chapter. When you are finished, answer the questions and work the problems given below. Place a mark beside each problem or question that you cannot answer, and then go back to the text and reread appropriate sections.

QUESTIONS

1. Define **transducer** in your own words.
2. List several electrical parameters that can be used to transduce physical parameters.
3. Many transducers are often used in ____________ bridge circuits.
4. Define **piezoresistivity** in your own words.

5. **Tension** forces in a cylindrical conductor ______________ the electrical resistance.
6. **Compression** forces in a cylindrical conductor ______________ the electrical resistance.
7. Changes in electrical resistance due to deformation are called ______________.
8. Define **gage factor** in your own words.
9. Write the two equations for gage factor.
10. Discuss the differences between **bonded** and **unbonded** strain gages.
11. Write the equation for E_0 in a Wheatstone bridge transducer that uses *one* active element.
12. What are the *units* for the sensitivity of the fluid pressure transducer?
13. Draw a circuit showing a **balance control** in a Wheatstone bridge strain gage.
14. List three types of temperature transducers.
15. Define **Seebeck effect** and **Peltier effect** in your own words. How are these effects related?
16. The temperature coefficients for most elemental metals are ______________.
17. Specially "doped" semiconductor materials can have both ______________ and ______________ temperature coefficients.
18. A thermistor used as an electronic thermometer must be operated to minimize ______________.
19. A thermocouple is formed of two ______________ metals and works because the metals have different ______________ ______________.
20. List three types of semiconductor temperature transducers.
21. List two types of inductive transducers. Draw appropriate circuit diagrams.
22. The secondary windings of an LVDT are connected in ______________.
23. List two types of position/displacement transducers.
24. Draw circuits for one- and two-quadrant position transducers.
25. How can a displacement transducer derive velocity and acceleration signals?
26. What types of devices are used to measure angular velocity?
27. List three types of phototransducers.
28. List two types of photoconductive semiconductor elements.

PROBLEMS

1. A constantan cylinder with a diameter of 0.05 mm is 14.6 cm long. Find its electrical resistance.
2. Predict the **gage factor** if a 12-mm-long strain gage element changes length 0.7 mm and its resistance changes from 78 to 82 Ω. What type of force is applied, **tension** or **compression?**
3. What is the gage factor of a cylindrical wire 0.05 mm in diameter and 18 mm long if the diameter changes 0.03 mm and the length changes 2.9 mm when a tension force is applied?
4. A Wheatstone bridge strain gage is excited by a 6-V dc source. When it is stimulated by an outside parameter, each element changes resistance by a factor of $0.042R$, and all four arms have equal resistance under zero-stimulus conditions. Calculate the output voltage E_0 for (a) one, (b) two, and (c) three active elements.

5. A 5-g mass hangs from the end of a bar-type strain gage (Fig. 13–12). Express this mass as a force in *dynes.*
6. A fluid pressure transducer has a sensitivity of 50 μV/V/cmHg and is excited by a +7.5-V dc source. Find the output voltage if a 150-mmHg pressure is applied.
7. The transducer in Problem 6 is a Wheatstone bridge strain gage, with four active elements that each have a zero-pressure resistance of 500 Ω. Calculate the value of a calibration resistance (to shunt one arm) that will give an artificial pressure of 120 mmHg.
8. A thermistor is known to have a resistance of 100 kΩ at 25°C and a temperature coefficient of $0.02°C^{-1}$. Calculate the resistance at (a) 30°C and (b) 56°C.
9. Find the resistance of a thermistor at 100°C if the ice-point resistance is 500 kΩ and the value of β is 2900 K.
10. Find the base-emitter voltage of a transistor at 33°C if the collector current is 10 mA and the reverse saturation current is 10^{-13} A.
11. Find the slope factor of a dual-transistor temperature transducer such as Fig. 13–9 if $I1$ = 10 mA and $I2$ = 3 mA.
12. What amplification following the transducer in Problem 11 results in a scale factor of 10 mV/K?

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